



TWP4213

Internet and Web Publishing

Assignment

Build a Responsive Admin Homepage (Bootstrap)

In this assignment, you will be able to:

- Use Bootstrap 5's grid, components, and utilities to design responsive layouts.
- Create a basic admin page layout with navigation and content sections.
- Use Bootstrap components like Navbar, Cards, Table, and Buttons.

Bootstrap is one of the most popular front-end frameworks for building responsive and mobile-first websites. Developed by Twitter, it provides a collection of pre-designed HTML, CSS, and JavaScript components that help developers create modern, visually appealing, and consistent web interfaces quickly.

For this assignment, you will use Bootstrap 5 to design a responsive homepage for an online store. This will help you understand how to structure layouts, apply Bootstrap's grid system, and use its components effectively to create a professional-looking website without writing extensive custom CSS.

Requirements

To use Bootstrap, first you need to have a starter template. Copy the starter template code from this link: <https://getbootstrap.com/docs/5.0/getting-started/introduction/#starter-template>.

1. Sidebar

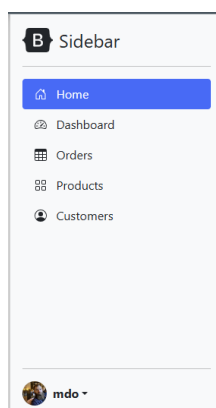
A sidebar is a vertical navigation panel that usually appears on the left or right side of a webpage. It's commonly used in admin dashboards or web apps to provide quick access to different sections or pages.

A sidebar keeps navigation **visible** while working on the main content and it saves space compared to a top navbar when there are many links.

Refer to this link for sample code and tutorial on how to create a Bootstrap sidebar.

<https://getbootstrap.com/docs/5.3/components/navs-tabs/#vertical>
[Sidebars · Bootstrap v5.3](#)

Example of a navigation bar.



2. Main content

Main content refers to the primary area of a webpage where the core information or functionality is displayed.

In an admin page layout with a sidebar:

The sidebar is for navigation (links to other sections). The main content is the section where the actual dashboard elements appear, such as:

- Stat cards (showing totals or summaries)
- Tables (listing records or data)
- Forms or modals (for adding/editing items)

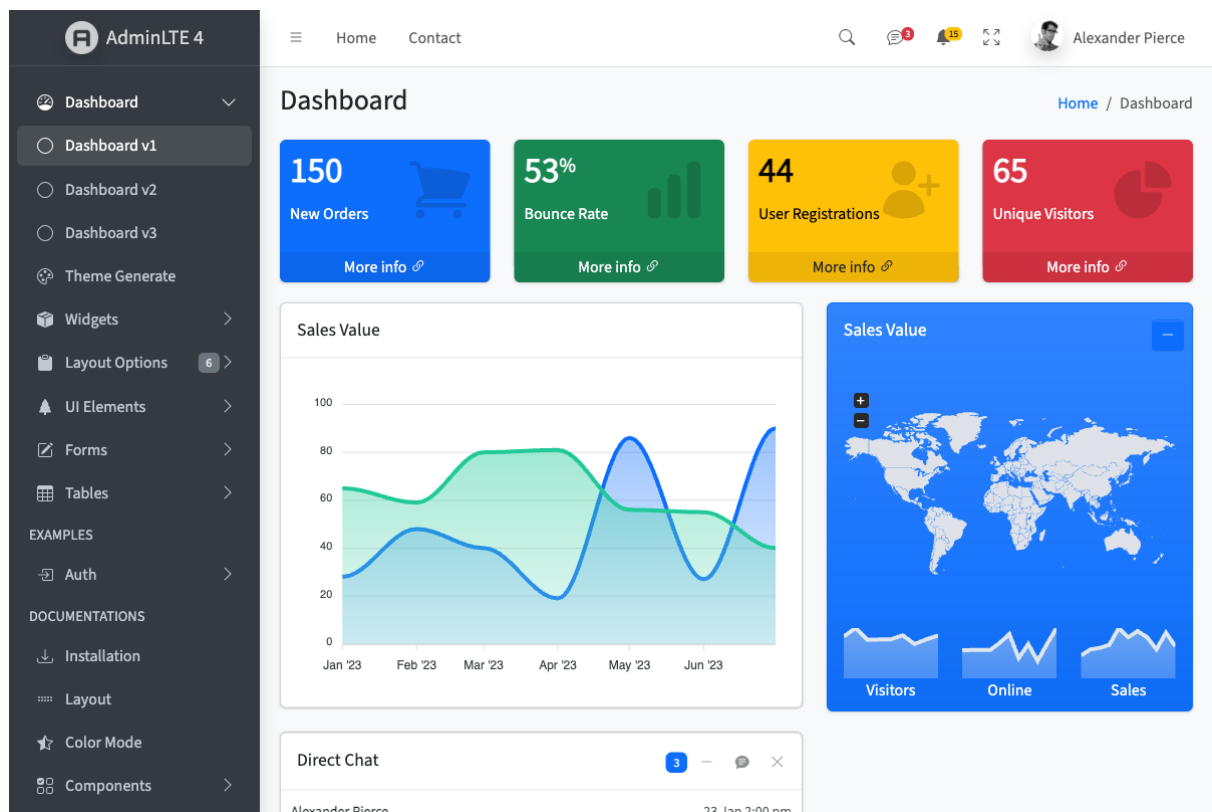
Think of it as the workspace where users interact with the system, while the sidebar is just the menu.

[Cards · Bootstrap v5.3](#)

[Tables · Bootstrap v5.3](#)

[Dashboard Template · Bootstrap v5.3](#)

Example of a main content.



3. Responsiveness

Responsiveness means your webpage adapts to different screen sizes (desktop, tablet, mobile) without breaking the layout or making content hard to read. Responsive design ensures good user experience across devices.

You can use bootstrap's grid system and utility class to achieve responsive design.

3.1. Use the Grid System

- Wrap content in `.container` or `.container-fluid`.
- Use `.row` for horizontal grouping.
- Use **responsive column classes** like:
 - `.col-12` for full width on mobile
 - `.col-md-6` for half width on medium

3.2. Responsive Sidebar

- On large screens, sidebar stays visible.
- On small screens, sidebar collapses and toggles with a button. Bootstrap's **collapse component** helps:

3.3. Responsive Table

- Wrap tables in `.table-responsive` so they scroll horizontally on small screens:

3.4. Responsive Utilities

- Use spacing classes like `.p-3`, `.m-2` for consistent padding/margin.
- Use display utilities like `.d-none` `d-lg-block` to hide/show elements based on screen size.

[Grid system · Bootstrap v5.3](#)

[Display property · Bootstrap v5.3](#)

[Tables · Bootstrap v5.3](#)

[Collapse · Bootstrap v5.3](#)

4. Styling

Styling refers to how your webpage looks and feels—its colors, spacing, typography, and visual hierarchy. Good styling makes the page readable, attractive, and consistent.

Styling improves user experience and readability, and it makes the layout look professional and organized.

4.1. Use Bootstrap Classes for Styling

- Apply Bootstrap's built-in classes for:
 - **Colors:** .bg-light, .text-primary, .btn-primary
 - **Spacing:** .p-3 (padding), .m-2 (margin)
 - **Typography:** .fw-bold (bold), .text-muted (lighter text)

4.2. Consistent Layout

- Use .container or .container-fluid for proper alignment.
- Use Bootstrap's **grid system** for responsive spacing.

4.3. Custom styling

- Change colours and theme by overriding the using internal CSS.
- Add branding elements like logo
- Modify default layout with your own setting.

[Colors · Bootstrap v5.3](#)

[Spacing · Bootstrap v5.3](#)

[Typography · Bootstrap v5.3](#)

[Buttons · Bootstrap v5.3](#)

Submission

- Students need to upload and attach in the Assignment in Teams a single **index.html** file that fulfil all the requirements.
- Make sure any customization done to default CSS is done as internal CSS.
- Custom CSS code need to include comment of what kind of changes are being applied.
- If there is any image used in the page than the index.html and images are compressed to a single .zip file before being uploaded.
- The submission deadline is 31st January 2026.

Evaluation and Marks

Criteria	Very Bad (1)	Bad (2)	Satisfactory (3)	Good (4)	Excellent (5)
Sidebar Navigation	Missing or broken	Sidebar present but poorly structured	Sidebar with basic links, no responsiveness	Sidebar with links and basic responsiveness	Sidebar fully functional, responsive, and styled
Main Content Layout	No main content	Minimal content, poor alignment	Cards and table present but basic	Cards and table well-aligned, responsive	Professional layout with cards, table, and good spacing
Responsiveness	Not responsive	Layout breaks on small screens	Works on mobile but poorly styled	Responsive and readable on all devices	Fully responsive with smooth sidebar toggle
Styling & Branding	No customization	Minimal changes, looks generic	Some custom colors/fonts	Good color scheme and branding	Unique, polished design with custom theme and fonts